

Pod Networking in Kubernetes

Since you're learning **KCNA as a beginner**, let's focus only on **Pod Networking**.

1. What is Pod Networking?

Pod Networking is the system that allows Pods to communicate with other Pods and with other network endpoints.

In simple words:

 **Pod Networking = How Pods talk to each other.**

2. Every Pod Gets an IP Address

When Kubernetes creates a Pod, the Pod gets an **IP address**.

Example:

Frontend Pod

IP: 10.1.0.5

Backend Pod

IP: 10.1.0.6

They can communicate:

Frontend Pod

10.1.0.5

|

| 

▼

Backend Pod

10.1.0.6

 **Remember:**

Pod = gets its own IP address

3. Pod-to-Pod Communication

One of the main goals of Kubernetes networking is:

Pods should be able to communicate with other Pods.

Example:

Pod A

|

| HTTP request



Pod B

Pod A doesn't need to know whether Pod B is on the same Node or another Node.

4. Pods on the Same Node

Suppose:

Node 1

|

|— Pod A

| IP: 10.1.0.5

|

|— Pod B

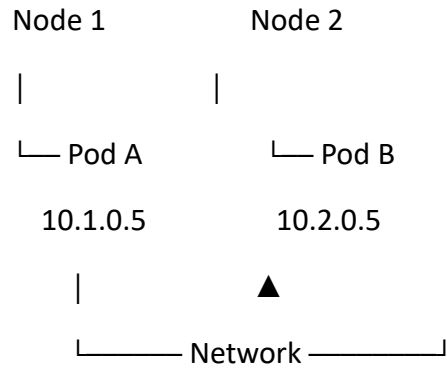
IP: 10.1.0.6

They can communicate through the Pod network:

Pod A —————→ Pod B

5. Pods on Different Nodes ★

This is especially important.



Pod A should be able to communicate with Pod B even though they are on different Nodes.
The Kubernetes networking implementation provides this connectivity.

6. Who Provides Pod Networking?

Kubernetes uses a **CNI (Container Network Interface)** implementation for Pod networking.

Examples of CNI/networking solutions include:

- **Cilium**
- **Calico**
- **Flannel**

Very simply:

Pod Created



CNI / Network Plugin



Pod gets network connectivity



Pod can communicate 

 **Easy memory:**

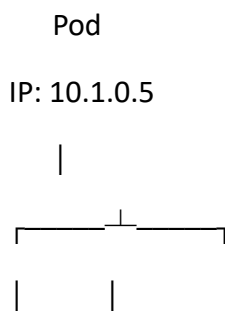
CNI = Provides/configures Pod networking

7. Containers Inside the Same Pod

This is very important.

Containers in the **same Pod share the Pod's network namespace and IP address.**

Example:



Container A Container B

They can communicate using:

localhost

For example:

Container A → localhost → Container B

provided they use different ports.

 **Remember:**

Same Pod = Same network/IP

8. Pod Networking vs Container Networking

Don't confuse these.

Container

Container

is the application environment.

Pod

Pod

└─ Container

└─ Container

Kubernetes assigns networking at the **Pod level**.

So:

Kubernetes gives the network identity/IP to the Pod, not separately to each container inside the Pod.

9. Pod Networking and Services

Pod IPs can change when Pods are recreated.

For example:

Old Pod

10.1.0.5



Deleted ❌

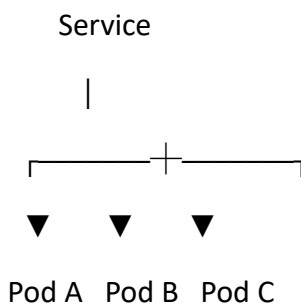
New Pod:

New Pod

10.1.0.20

So applications shouldn't depend on a specific Pod IP.

That's why Kubernetes provides **Services**.



The Service provides a stable way to reach a group of Pods.

10. Pod Networking + NetworkPolicy

These two concepts are related but different.

Pod Networking

Allows communication.

Pod A —————→ Pod B

NetworkPolicy

Controls/restricts communication.

Pod A —  —→ Pod B

or:

Pod A —  —→ Pod B

Easy memory:

 **Pod Networking = Communication**

 **NetworkPolicy = Control of communication**

11. Simple Real-Life Example

Think of Pods as houses.

Each house has an address:

House A → 10.1.0.5

House B → 10.1.0.6

House C → 10.1.0.7

The road system allows houses to communicate.

That's like:

Pod Networking

Security gates decide who is allowed to enter.

That's like:

NetworkPolicy

And a reception desk gives you a stable place to find a group of houses.

That's similar to:

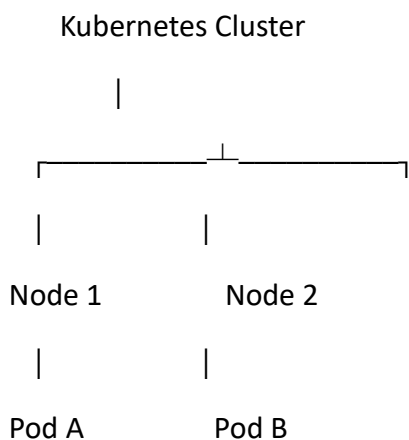
Service

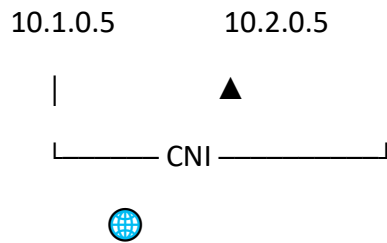
★ KCNA Revision Notes

Pod Networking

- Every Pod gets its own **IP address**.
 - Pods can communicate with other Pods.
 - Pods can communicate even when they are on different Nodes.
 - **CNI/network plugins** provide Pod networking.
 - Containers inside the same Pod share the **Pod's network namespace and IP**.
 - Containers in the same Pod can communicate using localhost.
 - Pod IPs can change when Pods are recreated.
 - **Services** provide a stable way to access Pods.
 - **NetworkPolicy** controls which network traffic is allowed.
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🧠 Super Easy Diagram





Pod Networking

Remember these 4 lines:

📌 Pod gets IP

🌐 Pod can communicate

🔑 CNI provides networking

🔒 NetworkPolicy controls traffic

One-line KCNA definition:

Pod Networking is the networking system that gives Pods network connectivity and enables communication between Pods within a Kubernetes cluster.